

Replication Report: OSEP Double-Well Demonstration in *First-principles studies of multiferroic and magnetoelectric materials*

(Fang et al., 2015; arXiv:1508.07414)

Automated replication (texture class: orbital)

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1 Nature of the target

The paper is a **review article** surveying first-principles (DFT) methods for multiferroic and magnetoelectric materials. It has no single crisp research result; instead it introduces the authors' *orbital selective external potential* (OSEP) method and demonstrates it on representative oxides. Because a full plane-wave DFT reproduction is out of scope (DFT-heavy, cluster-class), we target the paper's cleanest *machine-checkable* sub-result: the OSEP double-well demonstration for BaTiO₃ (BTO, Fig. 1) and PbTiO₃ (PTO, Fig. 2), Section 3.1.1.

2 Central claims replicated

- C1** BTO has a symmetric *double-well* total-energy curve vs. the Ti off-center displacement (A_{2u} soft mode) $\Rightarrow$ ferroelectric ground state.
- C2** The OSEP method shifts *only* the Ti-3d on-site energy. Shifting Ti-3d *up* weakens Ti-3d/O-2p hybridization and lowers the double well; at a shift of ~ 2 eV the double well *vanishes* (single well) $\Rightarrow$ ferroelectricity destroyed.
- C3** The polar instability is driven by Ti-3d/O-2p *hybridization* (second-order Jahn–Teller / Cohen mechanism): increasing the 3d–2p gap weakens the instability.
- C4** In PTO, shifting Ti-3d *down* or Pb-6s *up* both *deepen* the well, but the Pb-6s effect is *much weaker* than the Ti-3d effect.

3 Minimal model

We do not perform DFT. We build the minimal electronic model that produces a double well by exactly the mechanism the paper invokes. A single soft-mode coordinate Q (Ti off-center displacement, Å) couples a filled O-2p level to an empty Ti-3d level through a 2×2 vibronic Hamiltonian

$$H(Q) = \begin{pmatrix} \varepsilon_p & gQ \\ gQ & \varepsilon_d \end{pmatrix}, \quad E(Q) = \frac{1}{2}kQ^2 + k_4Q^4 + 2[\lambda_-(Q) - \lambda_-(0)],$$

where λ_- is the (occupied) lower eigenvalue, k the lattice stiffness (short-range repulsion favoring the cubic state), and k_4 an anharmonic bounding term. The cubic state is unstable (double well) when the net harmonic curvature

$$\left. \frac{d^2 E}{dQ^2} \right|_0 = k - \frac{4g^2}{\varepsilon_d - \varepsilon_p} < 0,$$

i.e. the vibronic softening ($\propto g^2/\Delta$, $\Delta = \varepsilon_d - \varepsilon_p$) beats the lattice stiffness — precisely the second-order Jahn–Teller condition the paper cites. OSEP is modeled by $\varepsilon_d \rightarrow \varepsilon_d + s$; the well vanishes at $s_{\text{crit}} = 4g^2/k - \Delta$.

Parameters were calibrated to physical BTO scales: bare gap $\Delta = 2.6$ eV (consistent with the Ti-3d conduction feature ~ 3 eV above E_F in the paper’s Fig. 1a PDOS), $g = 2.05$ eV/Å, $k = 3.654$ eV/Å² (fixes $s_{\text{crit}} = 2.0$ eV), $k_4 = 45$ eV/Å⁴ (gives a ~ 10 meV well at $|Q^*| \approx 0.12$ Å). PTO adds a weaker Pb-6s lone-pair channel.

4 Results

Claim	Quantity	Model	Paper / expectation
C1	BTO well depth; $ Q^* $	10.5 meV; 0.122 Å	double well; ~ 0.1 Å
C2	OSEP critical up-shift	2.00 eV (analytic), 1.69 eV (grid)	~ 2 eV
C2	depth vs shift (meV)	10.5, 4.2, 1.4, 0.27, 0 at 0, 0.5, 1, 1.5, 2 eV	monotone $\rightarrow 0$
C3	corr(depth, $1/\Delta$)	0.999	positive (hybridization)
C4	PTO deepen: Ti-3d down / Pb-6s up	28.3 / 2.95 meV	Ti $\gg$ Pb, both > 0

All four claims reproduce qualitatively and (for C2) quantitatively. The monotone collapse of the double well with OSEP up-shift and its disappearance at ~ 2 eV directly mirror Fig. 1b; the Ti-3d dominance over Pb-6s in PTO mirrors Fig. 2. Figures: `work/fig_bto_doublewell.png`, `work/fig_bto_welldepth_vs_shift.png`.

5 Verdict

Verdict: REPLICATED (mechanism-level, tractable surrogate). The paper’s OSEP demonstration is reproduced by a minimal second-order Jahn–Teller vibronic model that captures the same physics (hybridization-driven soft-mode instability) and matches the headline number (double-well quench at ~ 2 eV OSEP shift). This is a mechanism replication, not a DFT reproduction: the absolute well depth is model-calibrated, not first-principles.

Coverage: 6/10. We cover the one crisp quantitative sub-result of a broad review; the vast majority of the review (BFO, spin-spiral multiferroics, magnetoelectric interfaces, tunnel junctions) is descriptive/DFT-heavy and out of scope.

Agreement: 8/10. The replicated sub-result matches the paper’s qualitative trends exactly and its one quantitative claim (2 eV quench) on the nose; deduction because well depths are calibrated (the paper reports no absolute meV value to check against) and the grid-based critical shift lands at 1.69 eV.